Singapore Embodied AI Challenge for Manufacturing (ECM 2027) — Simulation Track Policy Submission

Team: TwinTeleop — **Team ID: ECM-122 Team lead:** Toni Adreal — toniadreal11@gmail.com **Submission date:** 2026-11-13 (Beijing) **Policy version:** v3.4 — multi-demo replay-follow (gripper replay + lagged-follower evaluation + lookahead sweep; submission default lookahead 2)

Where the code lives

Full policy code, tests, and this document are public at:

https://github.com/ToniAdreal/ecm2027-chamber-policy

Commit submitted: `1c4deaf2e8c17c54be7838c873b8af303cd33a50` ("ECM 2027 M8: lookahead sweep (6 demos x alpha 0.3/0.5 x lookahead 0/1/2/4/8, 62 tests green)", 2026-11-13)

This file is the submission README. The organizers asked for a policy file with a detailed README; because the submission form's upload validator rejects archive contents with code, data, and markdown extensions, this single PDF carries the full README and points at the public repository above for the runnable code. No other file is needed to evaluate the policy: clone the repository at the commit above and follow §4.

1. What the policy does

Multi-demonstration trajectory following for the Galbot-Mobile dual-arm chamber task in the official simulation (`scene04_1.usd`).

- At reset, the policy picks the recorded demonstration whose first-frame

end-effector positions are nearest to the robot's observed initial positions. Selection happens only at reset, so a rollout never jumps between demonstrations mid-episode. Without an observation at reset the first demonstration is used.

- During the episode it tracks a phase index into the chosen demonstration

and advances it monotonically. When the caller supplies the robot's current end-effector positions it re-anchors the phase to the demonstration point nearest the current pose within a forward window (closed-loop phase tracking); otherwise it advances one demonstration step per policy step.

- It commands the demonstration pose at `phase + lookahead` for both

arms, as absolute differential-IK pose commands (`use_relative_mode=False`, the starter repository's own convention).

- On Batch 2 demonstrations it replays the recorded gripper state:

`gripper_pos` is the Galbot gripper joint position (0.0 fully open, 1.703 fully closed), thresholded at the range midpoint 0.8515 into the binary gripper action (`< 0` close, `>= 0` open), the starter repository's own `BinaryJointPositionAction` convention. Batch 1 demonstrations carry no gripper channel, so on those the grippers are held at a fixed configurable command. Wheel velocity is commanded zero — see the honest limitation in §5.

The bundled set holds six real episodes: four Batch 1 (`demo_0`, `demo_2`, `demo_7`, `demo_9`) and two Batch 2 (`demo_b2_0`, `demo_b2_3`), extracted from six independent recordings of the official dataset `rocochallenge2025/ecm2027_sim_data`. Episode lengths: 7,471 / 5,632 / 7,566 / 5,122 / 4,106 / 4,022 steps. Batch 2 files carry the split observation terms `obs/eef_pos` + `obs/eef_quat` (composed into the same `(T, 2, 7)` shape by the loader) plus a real gripper observation `obs/gripper_pos` `(T, 2)`, stored in the `demo_b2_*` bundles.

2. Interfaces

Two adapters share the same core (`ecm_policy/core.py`, pure NumPy; multi-demo selection in `ecm_policy/multidemo.py`):

1. **Docker policy service** (`ecm_policy/service.py`) — the planned official submission shape, implementing the starter README's service operations `health()`, `reset(task_info)`, `predict(observation)`, `close()`. Defaults to multi-demo mode over `data/`; `--trajectory <single .npz>` selects the old single-demo mode. 2. **In-process adapter** (`ecm_policy/adapter.py`) — builds a `MultiDemoCore` when `trajectory_dir` / `demo_dir` / `ECM_TRAJECTORY_DIR` is set or when `trajectory_path` points at a directory, for use with the starter repository's own evaluation harness.

A `Dockerfile` is included in the repository. The starter README still lists the official Docker service transport and message schemas as pending; the service follows the operation names the starter README publishes.

3. Verification (all CPU-side; this host has no GPU)

- **62 / 62 tests pass** in the repository test suite, measured 2026-11-13

in an isolated clone at the submitted commit (57 pass under the system Python; the 5 Batch 2 layout tests need h5py and pass under the data-sample venv Python, where the full suite runs 62/62), including: full-episode rollout integrity (finite 20-dim actions, unit-norm quaternions, replayed gripper state on Batch 2 demos and fixed gripper on Batch 1

demos, zero wheels throughout, monotonic phase to trajectory end, no mid-episode demo switch), multi-demo selection, adapter, service, and core tests.

• Offline lagged-follower evaluation (`ecm_policy/evaluate.py`, full

table in `evaluation/results.md`): a simulated robot that moves a fraction `alpha` of the way toward each commanded pose per step. At `alpha = 1.0` the policy reproduces every bundled recording exactly (zero position error, zero phase lag); max phase lag strictly decreases as the follower gets faster on all 6 demos; gripper agreement on Batch 2 rises with `alpha` (32–42% at `alpha = 0.1` to 99.8–99.9% at `alpha = 1.0`). At `alpha` 0.1–0.5 the lagging follower cannot keep wall-clock pace (mean position error 0.50–0.70 m) — reported as a baseline, not a score.

• Lookahead sweep (`tools/sweep_lookahead.py`, full table in

`evaluation/lookahead_sweep.md`): every bundled demo at `alpha` 0.3 / 0.5 for lookaheads 0 / 1 / 2 / 4 / 8. Lookahead 2 is the submission default: it beats lookahead 1 at both swept alphas on demo-averaged mean position error (0.609 vs 0.618 m at `alpha = 0.3`; 0.010 vs 0.618 m at `alpha = 0.5`), while lookahead 4 wins only at 0.3 and lookahead 0 stalls the phase tracker outright (final phase 0). The near-perfect `alpha = 0.5` / lookahead-2 result is a property of the harness's first-order lag model, not a measurement of real Isaac Sim dynamics, and is reported as such.

• The starter repository's 7 supplied lightweight regression tests also

pass on this host (no simulator needed).

• **No score is claimed.** The policy has not been run in Isaac Sim on

this host (no GPU; Isaac Sim 6.0.1 needs an RTX GPU) and has not been evaluated by the organizers. The numbers above are test-suite results only.

4. How the organizers run it

```
 git clone https://github.com/ToniAdreal/ecm2027-chamber-policy.git
cd ecm2027-chamber-policy
git checkout 1c4deaf2e8c17c54be7838c873b8af303cd33a50
pip install -r requirements.txt   # numpy only
python3 -m pytest tests/ -q      # expect: 57 passed, 5 skipped (the 5
                                 # Batch 2 layout tests need h5py;
                                 # with h5py installed: 62 passed)
docker build -t ecm2027-policy . # policy service image
```

Demonstration data is bundled under `data/` (`demo_*_ee_poses.npz`, end-effector poses, shape `(T, 2, 7)` = `[x, y, z, qw, qx, qy, qz]` per arm; the `demo_b2_*` bundles also carry the recorded `gripper_pos` `(T, 2)`). The extraction tool is under `tools/`; Batch 2

data was released 2026-11-10 and is integrated in this package (two episodes); Batch 3 data (due 2026-11-28) can extend the set with the same tool.

5. Honest limitations

• **Gripper replay is built and tested; the mapping convention is

unverified by the organizers.** Batch 1 recordings carry no gripper channel, verified across all four independent Batch 1 recordings (each file contains exactly one `data/demo_*/ee_poses` dataset and nothing else), so Batch 1 cores still hold the fixed command. Batch 2 recordings carry `data/demo_*/obs/gripper_pos (T, 2)`, a real varying signal (0 to 1.703 in the two verified episodes), now consumed by the policy: full-episode open-loop rollouts replay the recorded gripper state exactly (demo_b2_0: 8 open/close transitions, 58.1% closed; demo_b2_3: 6 transitions, 68% closed). The mapping is the starter repository's own convention (`galbot.py`: gripper joint open/close positions 0.0 / 1.703; `action < 0` closes, `>= 0` opens), but the official evaluation service has never published its expected gripper action scale, so this remains a documented convention, not a validated organizer claim.

• **Fixed-layout assumption, weakened but not removed.** Multi-demo

selection picks the closest recorded start layout instead of assuming one fixed demonstration, but replay still assumes the evaluation scene starts near a recorded layout — the same premise the starter repository's own replay tooling uses.

• **CPU-side development only.** All verification above ran without a

simulator. Organizer-side evaluation is the intended validation step; the team welcomes the evaluation feedback the organizers send within 5 working days and will iterate on it.

• **Resubmission.** The team understands resubmission is allowed before

the policy submission deadline of **2026-12-30 12:00 SGT** and will resubmit an improved package (for example with Batch 3 demonstrations) if evaluation feedback calls for it.

6. Upstream issue filed

While setting up the starter repository the team found and reported a missing-module defect in the official starter code: https://github.com/rocochallenge/ecm2027_sim_track/issues/1 (`isaaclab_teleop` present in source but not installed; also missing `mdp` exports). Open with no organizer reply as of 2026-11-04.

Team TwinTeleop (ECM-122) — Toni Adreal, toniadreal11@gmail.com. Prepared 2026-11-06. Not yet uploaded to the submission form; the file is ready for the team lead's go-ahead.